

Chi-Square Test (χ^2 -Test)

The **Chi-Square Test** (χ^2 -Test) is a statistical tool primarily used to examine the relationship between two **categorical variables** or to compare an observed distribution with a theoretical one.



Parametric vs. Nonparametric Tests

Statistical hypothesis tests are broadly divided into two categories based on the assumptions they make about the population data:

Feature	Parametric Tests (e.g., <i>Z</i> -test, <i>t</i> -test, <i>F</i> -test)	Nonparametric Tests (e.g., χ^2 -test)
Data Type	Typically use interval or ratio data (measurable amounts).	Typically use nominal or ordinal data (categories or ranks).
Assumptions	Make specific assumptions about the population parameters: 1. Data follows a Normal Distribution . 2. Assumes Homogeneity of Variance .	Make fewer or no assumptions about the population distribution or variance. They are often called distribution-free tests.
Central Tendency	Based on mean (μ).	Often based on median or mode (counts/frequencies).
Chi-Square Test	χ^2 -test is Nonparametric .	

How the Chi-Square Test Statistic is Computed

The χ^2 test statistic is based on comparing the **Observed Frequencies** (O)—the counts actually recorded in the sample—with the **Expected Frequencies** (E)—the counts expected if the Null Hypothesis (H_0) were true.

Formula

The test statistic is the sum of the squared differences between the observed and expected counts, divided by the expected counts, across all categories or cells:

$$\chi^2_{\text{Calculated}} = \sum \frac{(O - E)^2}{E}$$

Where:

- O : The **Observed Frequency** (actual count) in each category/cell.
- E : The **Expected Frequency** (theoretical count) in each category/cell.

A **small** χ^2 value indicates that O and E are close, supporting H_0 . A **large** χ^2 value indicates a significant difference, leading to the rejection of H_0 .

Chi-Square Goodness-of-Fit Test

The **Goodness-of-Fit Test** determines if a sample distribution matches a hypothesized population distribution (e.g., checking if customer preferences are equally split among four flavors, or if student grades follow a known historical pattern).

Procedure

1. State Hypotheses:

- H_0 : The observed frequencies match the expected (theoretical) distribution.
- H_1 : The observed frequencies do not match the expected distribution.

2. Calculate Expected Frequencies (E):

E is calculated based on the H_0 assumption (e.g., if H_0 claims equal distribution among 5 categories and the total sample is 100, $E = 100/5 = 20$ for each category).

3. Calculate χ^2 :

Use the formula $\chi^2 = \sum \frac{(O-E)^2}{E}$.

4. Find Critical Value:

Use the χ^2 table with the chosen α and **Degrees of Freedom** ($df = \text{number of categories} - 1$).

5. Conclusion:

If $\chi^2_{\text{Calculated}} > \chi^2_{\text{Critical}}$, **Reject H_0** .



Chi-Square Test for Independence

The **Test for Independence** determines whether there is a statistically significant relationship between two **categorical variables** in a single population (e.g., is gender independent of voting preference?). The data is organized in a **contingency table**.

Procedure

1. State Hypotheses:

- H_0 : The two variables are **independent** (not related).
- H_1 : The two variables are **dependent** (related).

2. Calculate Expected Frequencies (E):

For each cell in the contingency table:

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

3. Calculate χ^2 :

Use the formula $\chi^2 = \sum \frac{(O-E)^2}{E}$ summed over all cells.

4. Find Critical Value:

- **Degrees of Freedom:** $df = (\text{Number of Rows} - 1) \times (\text{Number of Columns} - 1)$

5. Conclusion:

If $\chi^2_{\text{Calculated}} > \chi^2_{\text{Critical}}$, **Reject H_0** , concluding the variables are dependent.

Assumptions and Restrictions

The validity of the χ^2 -test relies on the following key conditions:

Assumptions

1. **Categorical Data:** The data must be in the form of **frequencies** (counts) from distinct categories.
2. **Random Sampling:** The sample must be drawn using a **random sampling** method.
3. **Independence of Observations:** Each observation must be independent of every other observation (e.g., one person cannot fall into two different cells of the table).

Restriction of Expected Frequency Size

The χ^2 approximation is only reliable if the sample size is large enough to ensure that the expected frequencies (E) are not too small.

- **Rule: No more than 20%** of the expected frequencies (E) should be less than **5**.
- **Absolute Minimum: Every** expected frequency (E) should be **at least 1**.

If this restriction is violated, the categories must be combined (pooled) to increase the expected cell counts, or a different test must be used.

The Chi-Square (χ^2) test uses observed counts to test two things: how well a sample distribution matches a theoretical distribution (Goodness-of-Fit) or whether two categorical variables are related (Test for Independence).

Hypothesis Testing (Chi-square test) Examples

Example 1: Chi-Square Goodness-of-Fit Test (Equal Frequencies)

Question: The demand for a spare part varied daily. Sample data: 1124, 1125, 1110, 1120, 1126, 1115 (for Mon-Sat). Expected frequency (E) is 1118 for each day. **Test the hypothesis.** (Assume $\alpha = 0.05$)

- Total Demand: $\sum O = 6720$ (The provided calculation of 1120 is incorrect; $6270/6 = 1045$ is also wrong. Using the data: $1124 + 1125 + 1110 + 1120 + 1126 + 1115 = 6720$).
- **Corrected Expected Frequency (E):** $E = 6720/6 = 1120$ for each of the 6 days.
- $n = 6$ categories.

1. Hypotheses and Critical Value

- H_0 : The demand is uniformly distributed across the six days (Observed = Expected).
- H_1 : The demand is not uniformly distributed.
- **Degrees of Freedom (df):** $df = \text{Categories} - 1 = 6 - 1 = 5$.
- **Critical Value (χ^2_{Critical}):** For $df = 5$ and $\alpha = 0.05$, $\chi^2_{\text{Critical}} \approx 11.07$.

2. Calculate the χ^2 Statistic

Day	Observed (O)	Expected (E)	$(O - E)$	$(O - E)^2$	$\frac{(O-E)^2}{E}$
Mon	1124	1120	4	16	$16/1120 \approx 0.0143$
Tue	1125	1120	5	25	$25/1120 \approx 0.0223$
Wed	1110	1120	-10	100	$100/1120 \approx 0.0893$
Thu	1120	1120	0	0	0
Fri	1126	1120	6	36	$36/1120 \approx 0.0321$
Sat	1115	1120	-5	25	$25/1120 \approx 0.0223$

Day	Observed (O)	Expected (E)	$(O - E)$	$(O - E)^2$	$\frac{(O-E)^2}{E}$
Total	6720	6720			$\sum = 0.1803$

$$\chi^2_{\text{Calculated}} = 0.1803$$

3. Conclusion

- $\chi^2_{\text{Calculated}} = 0.1803$
- $\chi^2_{\text{Critical}} = 11.07$
- Since $0.1803 < 11.07$, the calculated χ^2 falls within the acceptance region.
- **Result: Accept H_0** (Fail to Reject H_0).
- **Interpretation:** There is **no significant evidence** to suggest that the demand for the spare part varies significantly from day to day. The small differences observed are likely due to random chance.

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Example 2: Chi-Square Test for Independence (Ruler vs. Urban)

Question: Examine whether the **nature of area (Ruler/Urban)** related to the **voting preference** (A/B) in elections. (Assume $\alpha = 0.05$).

Area	Voter for A	Voter for B	Total
Ruler	$O = 620$	$O = 380$	1000
Urban	$O = 550$	$O = 450$	1000
Total	1170	830	2000

1. Hypotheses and Critical Value

- H_0 : Area and voting preference are **independent** (not related).

- H_1 : Area and voting preference are **dependent** (related).
- **Degrees of Freedom (df)**: $df = (\text{Rows} - 1) \times (\text{Columns} - 1) = (2 - 1) \times (2 - 1) = 1$.
- **Critical Value (χ^2_{Critical})**: For $df = 1$ and $\alpha = 0.05$, $\chi^2_{\text{Critical}} \approx 3.84$.

2. Calculate Expected Frequencies (E)

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

Cell (O)	Calculation (E)	Expected (E)
Ruler, Voter A (620)	$(1000 \times 1170)/2000$	585
Ruler, Voter B (380)	$(1000 \times 830)/2000$	415
Urban, Voter A (550)	$(1000 \times 1170)/2000$	585
Urban, Voter B (450)	$(1000 \times 830)/2000$	415

3. Calculate the χ^2 Statistic

Cell	O	E	$(O - E)$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
R, A	620	585	35	1225	$1225/585 \approx 2.0940$
R, B	380	415	-35	1225	$1225/415 \approx 2.9518$
U, A	550	585	-35	1225	$1225/585 \approx 2.0940$
U, B	450	415	35	1225	$1225/415 \approx 2.9518$
Total	2000	2000			$\Sigma = 10.0916$

$$\chi^2_{\text{Calculated}} = 10.0916$$

4. Conclusion

- $\chi^2_{\text{Calculated}} = 10.0916$
- $\chi^2_{\text{Critical}} = 3.84$
- Since $10.0916 > 3.84$, the calculated χ^2 falls into the rejection region.

- **Result: Reject H_0 .**
- **Interpretation:** There is a **significant relationship** between the nature of the area and voting preference. The variables are **dependent**.

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Example 3: Chi-Square Test for Independence (Inoculation Effectiveness)

Question: Use χ^2 test to defend or refute the statement that the **inoculation prevents attack from cholera**. (Assume $\alpha = 0.05$).

	Attacked (A)	Not Attacked (NA)	Total
Inoculated (I)	$O = 30$	$O = 160$	190
Not Inoculated (NI)	$O = 140$	$O = 460$	600
Total	170	620	790

1. Hypotheses and Critical Value

- H_0 : Inoculation and being attacked are **independent** (Inoculation has no effect).
- H_1 : Inoculation and being attacked are **dependent** (Inoculation prevents attack).
- **Degrees of Freedom (df):** $df = (2 - 1) \times (2 - 1) = 1$.
- **Critical Value (χ^2_{Critical}):** For $df = 1$ and $\alpha = 0.05$, $\chi^2_{\text{Critical}} \approx 3.84$.

2. Calculate Expected Frequencies (E)

Cell	Calculation (E)	Expected (E)
I, A	$(190 \times 170)/790$	40.9
I, NA	$(190 \times 620)/790$	149.1

Cell	Calculation (E)	Expected (E)
NI, A	$(600 \times 170)/790$	129.1
NI, NA	$(600 \times 620)/790$	470.9

3. Calculate the χ^2 Statistic

Cell	O	E	$(O - E)$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
I, A	30	40.9	-10.9	118.81	$118.81/40.9 \approx 2.905$
I, NA	160	149.1	10.9	118.81	$118.81/149.1 \approx 0.797$
NI, A	140	129.1	10.9	118.81	$118.81/129.1 \approx 0.920$
NI, NA	460	470.9	-10.9	118.81	$118.81/470.9 \approx 0.252$
Total	790	790			$\Sigma = 4.874$

$$\chi^2_{\text{Calculated}} = 4.874$$

4. Conclusion

- $\chi^2_{\text{Calculated}} = 4.874$
- $\chi^2_{\text{Critical}} = 3.84$
- Since $4.874 > 3.84$, the calculated χ^2 falls into the rejection region.
- **Result: Reject H_0 .**
- **Interpretation:** The statement is **defended**. There is a **significant relationship** between inoculation status and being attacked. Inoculation appears to be effective in preventing cholera attacks.



Quiz and Assignment Questions Solutions

Quiz 1: Chi-Square Goodness-of-Fit Test (Ratio)

Question: 500 students. Observed: 220 failed, 170 third, 90 second, rest first. Test if figures support the ratio **4:3:2:1**. ($\alpha = 0.05$)

1. Calculate Observed (O) and Expected (E) Frequencies

- **Total Sample:** $n = 500$.
- **Observed Frequencies (O):**
 - Failed: 220
 - Third: 170
 - Second: 90
 - First: $500 - (220 + 170 + 90) = 20$
- **Ratio Total:** $4 + 3 + 2 + 1 = 10$.
- **Expected Frequencies (E):**
 - Failed: $500 \times (4/10) = 200$
 - Third: $500 \times (3/10) = 150$
 - Second: $500 \times (2/10) = 100$
 - First: $500 \times (1/10) = 50$

2. Hypotheses and Critical Value

- H_0 : The categories are in the ratio 4 : 3 : 2 : 1.
- H_1 : The categories are not in the ratio 4 : 3 : 2 : 1.
- **Degrees of Freedom (df):** $df = 4 - 1 = 3$.
- **Critical Value (χ^2_{Critical}):** For $df = 3$ and $\alpha = 0.05$, $\chi^2_{\text{Critical}} \approx 7.815$.

3. Calculate the χ^2 Statistic

Category	O	E	$(O - E)$	$(O - E)^2$	$\frac{(O-E)^2}{E}$
Failed	220	200	20	400	$400/200 = 2.00$

Category	O	E	$(O - E)$	$(O - E)^2$	$\frac{(O-E)^2}{E}$
Third	170	150	20	400	$400/150 \approx 2.67$
Second	90	100	-10	100	$100/100 = 1.00$
First	20	50	-30	900	$900/50 = 18.00$
Total	500	500			$\Sigma = \mathbf{23.67}$

$$\chi^2_{\text{Calculated}} = \mathbf{23.67}$$

4. Conclusion

- $\chi^2_{\text{Calculated}} = 23.67$
- $\chi^2_{\text{Critical}} = 7.815$
- Since $23.67 > 7.815$, **Reject H_0** .
- **Answer:** The figures **do not support** the general belief that the categories are in the ratio 4 : 3 : 2 : 1.

Quiz 2: Chi-Square Test for Independence (Opinion vs. Class Level)

Question: 400 UG and 400 PG students. 290 UG favored, 310 PG favored. Test if **opinion is independent of class level**. ($\alpha = 0.05$).

1. Create Contingency Table and Hypotheses

Class Level	Favored (F)	Not Favored (NF)	Total
UG	$O = 290$	$400 - 290 = 110$	400
PG	$O = 310$	$400 - 310 = 90$	400
Total	600	200	800

- H_0 : Opinion is **independent** of class level.
- H_1 : Opinion is **dependent** on class level.

- **Degrees of Freedom (df):** $df = (2 - 1) \times (2 - 1) = 1$.
- **Critical Value (χ^2_{Critical}):** For $df = 1$ and $\alpha = 0.05$, $\chi^2_{\text{Critical}} \approx 3.84$.

2. Calculate Expected Frequencies (E)

Since row totals are equal (400) and column totals are 600 and 200:

- $E(G, F) = E(PG, F) = (400 \times 600)/800 = 300$
- $E(G, NF) = E(PG, NF) = (400 \times 200)/800 = 100$

3. Calculate the χ^2 Statistic

Cell	O	E	$(O - E)$	$(O - E)^2$	$\frac{(O-E)^2}{E}$
UG, F	290	300	-10	100	$100/300 \approx 0.333$
UG, NF	110	100	10	100	$100/100 = 1.000$
PG, F	310	300	10	100	$100/300 \approx 0.333$
PG, NF	90	100	-10	100	$100/100 = 1.000$
Total	800	800			$\Sigma = 2.666$

$$\chi^2_{\text{Calculated}} = 2.666$$

4. Conclusion

- $\chi^2_{\text{Calculated}} = 2.666$
- $\chi^2_{\text{Critical}} = 3.84$
- Since $2.666 < 3.84$, **Accept H_0** .
- **Answer:** Opinion regarding autonomous status of colleges is **independent** of the level of classes of students.

Quiz 3: Chi-Square Goodness-of-Fit (Poisson Distribution, Implicit)

Question: Number of defects per module in a sample of 330 units. Test the hypothesis.
(Given χ^2_{Critical} for 1 d.f = 3.84).

- **Note:** This problem typically tests if the observed defects follow a Poisson distribution, but without the mean () or formula, we can't perform that calculation. We will assume a simplified test, such as collapsing the low-frequency cells to satisfy the expected frequency restriction.
- **Given $df = 1$** implies the calculation required collapsing the data into two effective categories.

Simplified Approach (Collapsing Data):

Let's combine categories 2, 3, and 4 to meet the expected frequency restriction (though E values are not calculated here).

- **Observed Frequencies (O):** $O_1 = 214$ (0 defects), $O_2 = 92$ (1 defect), $O_3 = 20$ (2 defects), $O_4 = 13$ (3 defects), $O_5 = 4$ (4 defects).
- **Hypothetical Test:** Assume the null hypothesis was about whether the proportion of 0 defects vs. 1 or more defects matches a certain theoretical split, resulting in $df = 1$. Since we cannot derive E without the distribution's mean, we rely on the conclusion implied by the provided critical value.

Quiz 4: Chi-Square Test for Independence (Preference vs. Income)

Question: Test whether the **preference pattern of consumers for its product is dependent on income levels**. (χ^2_{Critical} for 4 d.f = 9.49).

Product Preferred	Low Income (LI)	Medium Income (MI)	High Income (HI)	Total
Product A (A)	170	50	20	240
Product B (B)	30	25	10	65
Product C (C)	80	60	55	195
Total	280	135	85	500

1. Hypotheses and Critical Value

- H_0 : Preference and income level are **independent**.
- H_1 : Preference and income level are **dependent**.
- **Degrees of Freedom (df)**: $df = (3 - 1) \times (3 - 1) = 4$.
- **Critical Value (χ^2_{critical})**: Given as **9.49**.

2. Calculate Expected Frequencies (E)

Cell	Calculation (E)	E
A, LI	$(240 \times 280)/500$	134.4
A, MI	$(240 \times 135)/500$	64.8
A, HI	$(240 \times 85)/500$	40.8
B, LI	$(65 \times 280)/500$	36.4
B, MI	$(65 \times 135)/500$	17.55
B, HI	$(65 \times 85)/500$	11.05
C, LI	$(195 \times 280)/500$	109.2
C, MI	$(195 \times 135)/500$	52.65
C, HI	$(195 \times 85)/500$	33.15

- **Check Restriction**: Only one cell (B, HI: 11.05) is below 5, which is acceptable.

3. Calculate the χ^2 Statistic

Cell	O	E	$\frac{(O-E)^2}{E}$
A, LI	170	134.4	9.68
A, MI	50	64.8	3.40
A, HI	20	40.8	10.51
B, LI	30	36.4	1.15
B, MI	25	17.55	3.16
B, HI	10	11.05	0.10
C, LI	80	109.2	7.74

Cell	O	E	$\frac{(O-E)^2}{E}$
C, MI	60	52.65	1.03
C, HI	55	33.15	14.88
Total	500	500	$\Sigma = 51.65$

$$\chi^2_{\text{Calculated}} = 51.65$$

4. Conclusion

- $\chi^2_{\text{Calculated}} = 51.65$
- $\chi^2_{\text{Critical}} = 9.49$
- Since $51.65 > 9.49$, **Reject** H_0 .
- **Answer:** We conclude that the preference patterns are **dependent on income levels**.